

Q1. Select the correct answer for the following questions:

1. The input-output relationship for LTI system $y(t) = \int_{-\infty}^t 2x(\tau - 3)d\tau$, then the impulse response of the system is:

- a. $y(t) = 2$ **b. $h(t) = 2\delta(t - 3)$** c. $h(t) = 2u(t - 3)$ d. $h(t) = 2t$ e. $y(t) = 0$

2. Which of the following system is memoryless?

- a) $h(t) = 0, t \neq 0$. **b. $h(t) = x(t - 1)$** . **c. $h(t) = 0, t = 0$** . d) $h(t) = kx(t + 2)$

3. The Fourier Transform of the signal $x(5t - 3)$ in terms of $X(j\omega)$ is given as:

- a. $\frac{1}{5}e^{-j\frac{3\omega}{5}}X(\frac{j\omega}{5})$** b. $\frac{1}{5}e^{j\frac{3\omega}{5}}X(\frac{j\omega}{5})$ c. $\frac{1}{5}e^{-j3\omega}X(\frac{j\omega}{5})$ d. $\frac{1}{5}e^{j3\omega}X(\frac{j\omega}{5})$

4. The impulse response $h(t)$ of linear time-invariant continuous time system is given by $h(t) = e^{-2t}u(t)$, then $H(\omega)$ equal

- a. $\frac{1}{1+2j\omega}$ b. $\frac{\sin \omega}{\omega}$ **c. $\frac{1}{2+j\omega}$** d. $\frac{j\omega}{2+j\omega}$

5. For a LTI system for an impulse response $h(t) = e^{2t}u(t - 1)$ is considered as....

- a. Causal. b. Memoryless. **c. Stable.** d. a and b. e. b and c f. Not mentioned.

6. Check the periodicity of the function $x(t) = \cos(2t) + \sin(3t)$. If it is periodic, the fundamental period is....

- a. The function is not periodic.
b. The function is periodic with a fundamental period of 5.2π
c. The function is periodic with a fundamental period of $\frac{5}{4}\pi$
d. The function is periodic with a fundamental period of 2π
e. d. The function is periodic with a fundamental period of $\frac{4}{5}\pi$

7. The given LTI system $y(t) = e^{-4t}u(t)$ is described as:

- a. Unstable and memoryless system.
b. Unstable and causal system.
c. Stable and causal system.
d. Memoryless and causal system.
e. Memoryless, causal and unstable system.

8. The Fourier Transform of $x(t) = \frac{d \operatorname{sgn}(t)}{dt}$ is given by:

- a. $X(\omega) = j\omega \delta(\omega)$ **b. $X(\omega) = 2$** c. $X(\omega) = \frac{2}{j\omega} + \pi\delta(\omega)$ d. $X(\omega) = \frac{2}{j\omega}$

9. Which of the following systems is time-invariant?

- a. $y(t) = x(\frac{-t}{2})$ b. $y(t) = x(t)(t - 2)$ c. $y(t) = x(1 - t)$ d. $y(t) = x(t) + x(t - 4)$

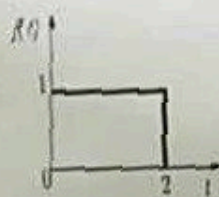
as figure given below, the $g(t)$ can be expressed as.....

a. $g(t) = f(2t - 3)$

b. $g(t) = f(\frac{t}{2} - 3)$

c. $g(t) = f(2t - \frac{3}{2})$

d. $g(t) = f(\frac{t}{2} - \frac{3}{2})$



20. The Fourier series expansion of real periodic signal with fundamental frequency f_0 is given by $g_p(t) = \sum_{n=-\infty}^{\infty} C_n e^{j2\pi n f_0 t}$. Given that $C_3 = 3 + j5$. The value of C_{-3} is

a. $5 + j3$

b. $-5 + j3$

c. $-3 - j5$

d. $3 - j5$

Q2. For a function of $x(t) = 1 + \sin(3\pi t) + \cos(4\pi t)$, answer the following:

2.1. Is this function periodic or not? periodic

2.2 If the function is periodic, what is the fundamental period?

$\sin(3\pi t) = T_1 = \frac{2\pi}{3\pi} \Rightarrow T_1 = \frac{2}{3}$

$\cos(4\pi t) = T_2 = \frac{2\pi}{4\pi} = \frac{1}{2}$

$\frac{T_1}{T_2} = \frac{\frac{2}{3}}{\frac{1}{2}} = \frac{4}{3}$

$T_0 = \frac{2}{3} \times 3 \Rightarrow T_0 = 2$

$T_0 = \frac{T_1}{2} \times 4 = 2$

2.3. Based on the answer in Q2.2, find the coefficients of $x(t)$ if they are available.

$C_0 = 1 + \frac{e^{j3\pi t} - e^{-j3\pi t}}{2j} + \frac{e^{j4\pi t} + e^{-j4\pi t}}{2}$

$1 + \frac{e^{j3\pi t}}{2j} - \frac{e^{-j3\pi t}}{2j} + \frac{e^{j4\pi t}}{2} + \frac{e^{-j4\pi t}}{2}$

$C_0 = 1, C_3 = -\frac{1}{2}j, C_{-3} = \frac{1}{2}j, C_4 = \frac{1}{2}, C_{-4} = \frac{1}{2}$



$C_3 = \sqrt{(\frac{1}{2})^2} = \frac{1}{2}$

$C_{-3} = \sqrt{(\frac{1}{2})^2} = \frac{1}{2}$

Good Luck

$$= -2\pi j[\delta(\omega - 20\pi) - \delta(\omega + 20\pi)], \text{ then } f(t) \text{ is expressed as:}$$

$$\pi \cos(\omega t)$$

$$= 2\pi \sin(\omega t)$$

$$= 2\pi \sin(\omega - 20\pi t)$$

$$f(t) = 2 \sin(20\pi t)$$

The z-transform of $f[n] = a^n u[n]$ is:

- a. $\frac{1}{z-a}$ b. $\frac{z}{(z-a)}$ c. $\frac{z}{(z-1)^2}$ d. $\frac{z}{z-1}$ e. None of them.

12. All causal systems must be a

- a. Time invariance b. Memory c. Linearity d. Both b and c e. Both b and d f. Not mentioned.

13. The Laplace Transform of $x(t) = \delta(t + 7)$ is given by:

- a. e^{7s} b. $\frac{1}{e^{-7s}}$ c. 7 d. πe^{-7s} e. None of them.

14. The Fourier transform of $\text{sinc}\left(\frac{t}{2}\right) * \delta(t)$ is:

- a. $\text{sinc}\left(\frac{\omega}{2}\right)$ b. $\text{sinc}^2\left(\frac{\omega}{2}\right)$ c. $\text{rect}\left(\frac{\omega}{2}\right) e^{\frac{j\omega}{2}}$ d. $2\pi \text{rect}(\omega)$

15. Which of the following functions can be represented by a Fourier series?

- a. $x(t) = 3 \sin(5t) + \cos\left(3t + \frac{\pi}{6}\right) + 2$
 b. $x(t) = \sin(5t) + \sin(3\pi t)$
 c. $x(t) = t e^{-2t}$
 d. $x(t) = \cos(2t) - 4 \cos\left(\frac{5}{2}\pi t\right) + 2$

16. If $x_1(t) = \sin(6t)$ and $x_2(t) = \cos(2\pi t)$ then the signal $x(t) = x_1(t) + x_2(t)$ is described as:

- a. Periodic signal with fundamental period 2π .
 b. Non periodic signal.
 c. Even signal.
 d. Periodic signal with fundamental period $\frac{\pi}{3}$.

17. The Fourier coefficients of the exponential form of $f(t) = \cos(t) + 3 \sin(2t)$, are:

- a. $c_0 = 1, c_1 = 2, c_{-1} = -2, c_2 = 1.5j, c_{-2} = -1.5j$
 b. $c_0 = 0, c_1 = \frac{1}{2}, c_{-1} = \frac{1}{2}, c_2 = -1.5j, c_{-2} = 1.5j$
 c. $c_0 = 0, c_1 = \frac{3}{2}j, c_{-1} = -\frac{3}{2}j, c_2 = -1.5, c_{-2} = 1.5$
 d. $c_0 = 0, c_1 = \frac{1}{2}j, c_{-1} = -\frac{1}{2}j, c_2 = -1, c_{-2} = 1$

18. The power of the signal $y[n] = 4$ is ____

- a. 4 b. 12 c. 64 d. 16